30 January 1976
SUPERSEDING
MIL-C-16375F
18 June 1968

MILITARY SPECIFICATION

CLOTH, WIGAN, COTTON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for cotton wigan utilized in the manufacture of clothing items.

1.2 Classification. The cotton wigan cloth shall be of the following types as specified (see 6.2):

Type I	-	2.5 oz. per sq. yd.
Type II	-	3.6 oz. per sq. yd.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material. The cotton shall be carded, drawn and spun into singles yarn.

3.3 Color. The color of the finished cloth shall be either natural or Grey 342 as specified. When Grey 342 is specified (see 6.2), the color shall match the standard sample.

* 3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed cloth shall contain no more labile than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

3.3.2 Matching. The color of the dyed and finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

3.3.3 Colorfastness. The dyed cloth shall show colorfastness to laundering, perspiration and wet drycleaning equal to or better than the standard sample (see 6.3). When no standard sample has been established or designated as applicable to colorfastness, the dyed cloth shall show "fair" colorfastness to laundering and "good" colorfastness to perspiration and wet drycleaning when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Type	Weight per sq. yd. ounces (minimum)	Yarns per inch (minimum)		Breaking strength (pounds) (minimum)	
		Warp	Filling	Warp	Filling
I	2.5	40	34	35	25
II	3.6	48	40	52	28

3.4.1 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.4.2 Weave. The weave for both type I and type II shall be a plain weave.

* 3.5 Finish. The cloth shall have a firm, plain calender finish. Type I cloth shall have a warp stiffness of not less than 0.005 inch-pounds and not more than 0.015 inch-pounds when tested as specified in 4.4.

3.6 pH. The pH value of the water extract of the finished cloth shall be no lower than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.7 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 40 yards. Each length shall be put up in rolls as specified in PPP-P-1134.

3.8 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.9 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the point level specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.2.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of 4.2.2.1 through 4.2.2.4.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined, and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be one linear yard. The acceptable quality level shall be 4.0 defects per 100 units (yards). The sample size shall be in accordance with level I of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls selected shall be in accordance with table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

Defects

Hole, cut or tear
Smash
Knots 1/
Embedded crease or wrinkle
Two or more ends out for more than nine inches
Spots or stains

1/ To be scored when larger than limits shown on level D of Sears Fabric Defect Scales (see 6.4).

* 4.2.2.2 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in table II.

Defects

Width less than specified 1/
Overall uncleanness

1/ Shall not be scored unless it occurs for more than 5 yards per piece.

TABLE II. Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to and including 3200	3	0
3201 up to and including 10,000	5	0
10,001 up to and including 35,000	8	1
35,001 up to and including 150,000	13	2
150,001 and over	20	3

4.2.2.3 Examination for length.

4.2.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

4.2.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.2.4 Examination for compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking complies with the section 5 requirements.

4.4 Testing of the end item. The method of testing specified in FED-STD-191, wherever applicable and as listed in table II, shall be followed. The physical and chemical values specified in section 3 apply to the results of the determination made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be one continuous yard, full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Material (cotton yarn)	3.2	1200 1/
Labile sulfur	3.3.1	2020
Colorfastness to:		
Laundering	3.3.3	5610
Perspiration	3.3.3	5680
Wet drycleaning	3.3.3	5622

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

TABLE II. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4.2	Visual 2/
Stiffness, type I (warp direction)	3.5	5202
pH	3.6	2811

1/ Unless otherwise specified, a certificate shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made on each sample unit and the result reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

* 5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. Cloth covered by this specification is intended for use as interlining for clothing.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of cloth required (see 1.2).
(c) Color required (see 3.3).
(d) Width of cloth required (see 3.4.1).
(e) When length is other than specified (see 3.7).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

* 6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Co., Department 733Q. "Sears Defect Replica Kit", Sears Tower, Chicago, IL 60684.

6.5 The margins of this specification are marked with an asterisk to indicate where (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 82

Preparing activity:

Army - GL

Project No. 8305-0032

Review activities:

Army - MD
DSA - CT

User activity:

Navy - MC

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK ~~DEVELOPMENT CENTER~~ Research & Development Command
NATICK, MA 01760
OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



AMXNM-TD C

Commander
US Army Natick Development Center
ATTN: AMXNM-TD C
Natick, MA 01760

NATICK OP 1 MAY 75
FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-C-16375G CLOTH, WIGAN, COTTON

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:38Z
Check the source to verify that this is the current version before use.